

Chemistry unit 2

6.1 — Enthalpy Change and Standard Conditions

Definition of Enthalpy Change (ΔH)

- **Enthalpy Change (ΔH)** is the heat energy transferred in a chemical reaction at **constant pressure**.
- It is measured in **kJ mol^{-1}** .

Standard Conditions

- These are specific reference conditions to ensure consistency:
 - **Pressure** = 100 kPa
 - **Temperature** = 298 K (25°C)
 - **All substances** in their **standard states** (physical state at 100 kPa and 298 K).

Standard State

- The **standard state** of a substance is its most stable form under standard conditions.
 - E.g., $\text{O}_2(\text{g})$, $\text{H}_2(\text{g})$, $\text{H}_2\text{O}(\text{l})$, $\text{Na}(\text{s})$, $\text{Cl}_2(\text{g})$, $\text{C}(\text{s, graphite})$

6.2 — Exothermic vs Endothermic Reactions

Feature	Exothermic	Endothermic
ΔH sign	Negative ($\Delta H < 0$)	Positive ($\Delta H > 0$)
Heat flow	Heat is released to surroundings	Heat is absorbed from surroundings
Temperature	Increases in surroundings	Decreases in surroundings
Examples	Combustion, Neutralisation	Thermal decomposition, Photosynthesis

6.3 — Enthalpy Level Diagrams

Exothermic Reaction Diagram

- Reactants at a **higher energy level** than products.
- Downward arrow showing energy released.
- ΔH is **negative**.

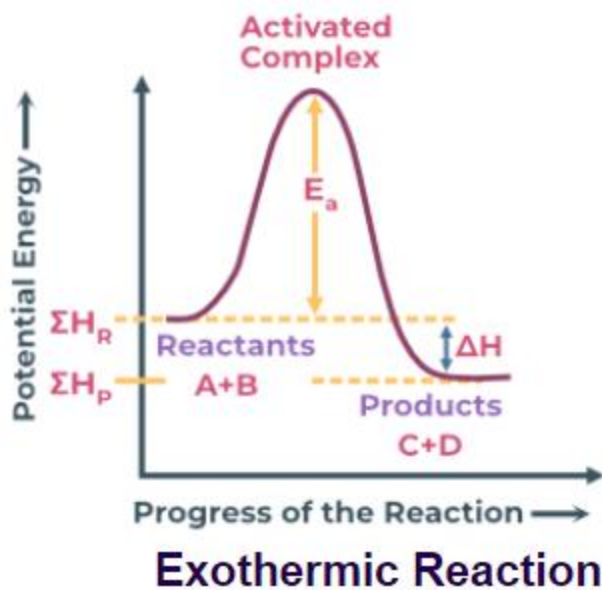
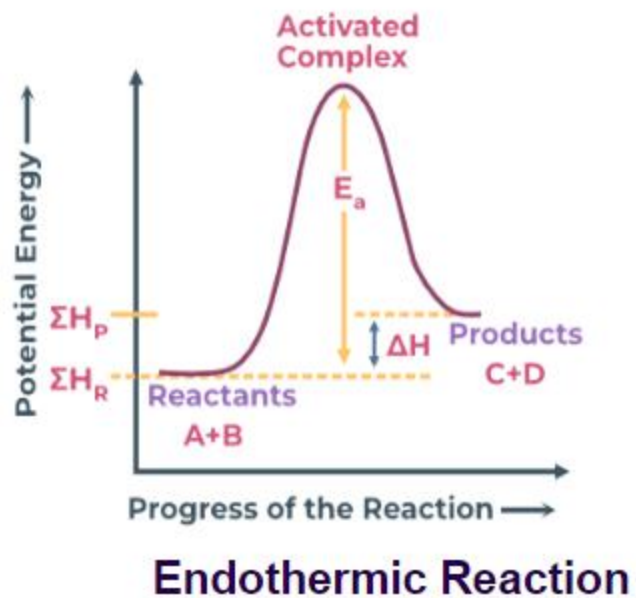
Endothermic Reaction Diagram

- Products at a **higher energy level** than reactants.
- Upward arrow showing energy absorbed.

- ΔH is **positive**.

Label axes:

- Y-axis: Enthalpy (H)
- X-axis: Progress of reaction



6.4 — Standard Enthalpy Changes

Term	Symbol	Definition	Example Equation	Type of Reaction
Standard Enthalpy Change of Reaction	$\Delta_r H^\circ$	Enthalpy change when molar amounts of reactants as shown in the balanced chemical equation react under standard conditions.	$\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$	Can be exothermic or endothermic
Standard Enthalpy Change of Formation	$\Delta_f H^\circ$	Enthalpy change when 1 mol of a compound is formed from its elements in their standard states under standard conditions.	$\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$	Can be exothermic or endothermic
Standard Enthalpy Change of Combustion	$\Delta_c H^\circ$	Enthalpy change when 1 mol of a substance is completely burned in excess oxygen under standard conditions.	$\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$	Always exothermic
Standard Enthalpy Change of Neutralisation	$\Delta_{\text{neut}} H^\circ$	Enthalpy change when an acid and a base react to form 1 mol of water under standard conditions.	$\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$	Always exothermic
Standard Enthalpy Change of Atomisation	$\Delta_{\text{at}} H^\circ$	Enthalpy change when 1 mol of gaseous atoms is formed from the element in its standard state.	$\frac{1}{2}\text{Cl}_2(\text{g}) \rightarrow \text{Cl}(\text{g})$	Always endothermic

Additional Important Enthalpy Terms (not in Unit 2 spec but useful)

Term	Symbol	Definition	Example Equation	Type of Reaction
Enthalpy Change of Solution	$\Delta_{\text{sol}} H^\circ$	Enthalpy change when 1 mol of an ionic compound dissolves in water to form a solution of infinite dilution.	$\text{NaCl}(\text{s}) \rightarrow \text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq})$	Can be exothermic or endothermic
Lattice Enthalpy (dissociation)	$\Delta_{\text{le}} H^\circ$	Enthalpy change when 1 mol of an ionic compound is separated into its gaseous ions.	$\text{NaCl}(\text{s}) \rightarrow \text{Na}^+(\text{g}) + \text{Cl}^-(\text{g})$	Always endothermic
Enthalpy of Hydration	$\Delta_{\text{hyd}} H^\circ$	Enthalpy change when 1 mol of gaseous ions become hydrated (i.e.,	$\text{Na}^+(\text{g}) \rightarrow \text{Na}^+(\text{aq})$	Always exothermic

		dissolved in water to become aqueous ions).		
Enthalpy of Sublimation	$\Delta_{\text{sub}}H^\circ$	Enthalpy change when 1 mol of a solid turns directly into a gas under standard conditions.	$\text{I}_2(\text{s}) \rightarrow \text{I}_2(\text{g})$	Always endothermic
Bond Dissociation Enthalpy	Δ^dH°	Enthalpy change required to break 1 mol of specific covalent bonds in gaseous molecules.	$\text{H}_2(\text{g}) \rightarrow 2\text{H}(\text{g})$	Always endothermic
Mean Bond Enthalpy	—	Average enthalpy required to break a particular bond in a range of compounds in gaseous state.	C–H bond $\approx +412 \text{ kJ mol}^{-1}$	Always endothermic
First Ionisation Energy	IE_1	Enthalpy change when 1 mol of gaseous atoms loses 1 electron to form 1 mol of gaseous 1^+ ions.	$\text{Na}(\text{g}) \rightarrow \text{Na}^+(\text{g}) + \text{e}^-$	Always endothermic
Electron Affinity	EA	Enthalpy change when 1 mol of gaseous atoms gains 1 electron to form 1 mol of gaseous 1^- ions.	$\text{Cl}(\text{g}) + \text{e}^- \rightarrow \text{Cl}^-(\text{g})$	1st EA is exothermic, 2nd EA is endothermic

Exam Tips

- Always state "**1 mol**" or "**molar amounts**" — this is often the difference between a 2/2 and a 1/2.
- Enthalpy of atomisation is **always positive** because it requires energy to break bonds.
- Combustion and neutralisation are **always exothermic**, even if the reaction “looks” small.
- Be careful with **formation** — not all are exothermic (e.g. formation of NO_2 is endothermic).

6.5 — Calculating Energy and Enthalpy Changes from Experiments

6.5(i) Energy Transferred in a Reaction (q)

Formula to use:

$$q = m \times c \times \Delta T$$

Symbol	Meaning	Units
q	Energy transferred	joules (J)
m	Mass of solution (or water) heated or cooled	grams (g)
c	Specific heat capacity of water	$4.18 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$
ΔT	Temperature change = Final temp – Initial temp	$^{\circ}\text{C}$ or K (same Δ)

Important Note:

- When mixing two solutions, **m = total mass of the liquid** (often assumed to have density of water: **1 g cm⁻³**, so $50 \text{ cm}^3 = 50 \text{ g}$).
- Use **c = $4.18 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$** unless otherwise specified.

Example Calculation of Energy Transferred

Reaction: Neutralisation of HCl and NaOH in a polystyrene cup

- 25.0 cm^3 of 1.00 mol dm^{-3} HCl reacts with 25.0 cm^3 of 1.00 mol dm^{-3} NaOH.
- The temperature rises from $20.0 \text{ }^{\circ}\text{C}$ to $26.5 \text{ }^{\circ}\text{C}$.

Step 1: Total volume = $50.0 \text{ cm}^3 \Rightarrow m = 50.0 \text{ g}$

Step 2: $\Delta T = 26.5 - 20.0 = 6.5 \text{ }^{\circ}\text{C}$

Step 3: $q = 50.0 \times 4.18 \times 6.5 = \mathbf{1,358.5 \text{ J}}$

6.5(ii) Enthalpy Change of Reaction (ΔH) in kJ mol^{-1}

Formula to use:

$$\Delta H = -q / n \quad (\text{convert } q \text{ to kJ first})$$

Symbol	Meaning	Units
ΔH	Enthalpy change	kJ mol^{-1}
q	Energy transferred	kJ
n	Moles of limiting reagent	mol

Signs:

- Negative ΔH = **exothermic**
- Positive ΔH = **endothermic**

Continue the Example

$$q = 1,358.5 \text{ J} = 1.3585 \text{ kJ}$$

$$\text{Moles of HCl} = 25.0 \div 1000 \times 1.00 = 0.025 \text{ mol}$$

(HCl and NaOH in 1:1 ratio $\Rightarrow$ both 0.025 mol, so either can be used)

$$\Delta H = -1.3585 \div 0.025 = -54.34 \text{ kJ mol}^{-1}$$

Final Answer: $\Delta H = -54.3 \text{ kJ mol}^{-1}$ (2/2 in exam)

Special Case: Combustion Experiments (using calorimeter)

- Metal can (copper) or bomb calorimeter used.
- Known mass of **fuel** is burned.
- **Water** absorbs the heat, temperature rise is measured.

Stepwise Approach:

1. Measure **mass of fuel burned**.
2. Measure **mass of water heated**.
3. Calculate $q = m \times c \times \Delta T$ for water.
4. Calculate **moles of fuel** = mass / Mr
5. Use $\Delta H = -q / n$

Potential Errors in Calorimetry Experiments

Source of Error	Effect on Results
Heat lost to surroundings	q too low $\Rightarrow \Delta H$ less exothermic than true value
Incomplete combustion (in fuel burns)	Less energy released $\Rightarrow \Delta H$ appears less exothermic
Evaporation of fuel/water	Mass lost inaccurately $\Rightarrow$ error in moles or q
Not stirring properly	Uneven heating $\Rightarrow$ temperature measured not accurate
Assuming density = 1.00 g cm^{-3} always	Not valid for non-aqueous solvents

6.6 — Hess's Law and Enthalpy Cycles

Definition

Hess's Law states:

The total enthalpy change for a chemical reaction is the same, no matter how the reaction takes place, provided the initial and final conditions are the same.

This means enthalpy is a **state function** — only the starting and ending points matter, not the path taken.

Types of Hess Cycles in Exams

There are three common cases:

1. Using enthalpy of formation ($\Delta_f H^\circ$)
2. Using enthalpy of combustion ($\Delta_c H^\circ$)
3. Using bond enthalpies

Case 1: Using Enthalpy of Formation

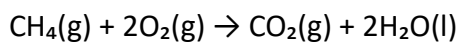
Equation type:

- Given: $\Delta_f H^\circ$ of all reactants and products
- Find: $\Delta_r H^\circ$

Hess formula:

$$\Delta_r H^\circ = \sum \Delta_f H^\circ(\text{products}) - \sum \Delta_f H^\circ(\text{reactants})$$

Example:



Given:

- $\Delta_f H^\circ(\text{CH}_4) = -75 \text{ kJ mol}^{-1}$
- $\Delta_f H^\circ(\text{CO}_2) = -394 \text{ kJ mol}^{-1}$
- $\Delta_f H^\circ(\text{H}_2\text{O}) = -286 \text{ kJ mol}^{-1}$
- $\Delta_f H^\circ(\text{O}_2) = 0$ (element in standard state)

Calculation:

$$\begin{aligned}\Delta_r H^\circ &= [-394 + 2(-286)] - [-75 + 0] \\ &= [-394 - 572] - [-75] = -966 + 75 = \mathbf{-891 \text{ kJ mol}^{-1}}\end{aligned}$$

Case 2: Using Enthalpy of Combustion

Equation type:

- Given: $\Delta_c H^\circ$ of all reactants and products
- Find: $\Delta_r H^\circ$ of another reaction

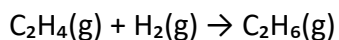
Construct a Hess Cycle:

- Arrows point **downwards** to combustion products (e.g., CO_2 and H_2O)
- Apply Hess's Law:

$$\Delta_r H^\circ = \sum \Delta_c H^\circ(\text{reactants}) - \sum \Delta_c H^\circ(\text{products})$$

Example:

Find $\Delta_r H^\circ$ for:



Given:

- $\Delta_c H^\circ(\text{C}_2\text{H}_4) = -1411 \text{ kJ mol}^{-1}$
- $\Delta_c H^\circ(\text{H}_2) = -286 \text{ kJ mol}^{-1}$
- $\Delta_c H^\circ(\text{C}_2\text{H}_6) = -1560 \text{ kJ mol}^{-1}$

Calculation:

$$\begin{aligned}\Delta_r H^\circ &= [-1411 + (-286)] - [-1560] \\ &= (-1697) - (-1560) = \mathbf{-137 \text{ kJ mol}^{-1}}\end{aligned}$$

6.8 — Evaluating Experimental Results and Commenting on Errors

A. Common Assumptions in Enthalpy Experiments

Assumption	Why It's Not Always Valid
No heat loss to surroundings	Heat escapes even with insulation; leads to less exothermic result
All solutions have the same specific heat capacity as water ($4.18 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$)	Ionic solutions may differ
Density of solution = 1.00 g cm^{-3}	Only accurate for dilute aqueous solutions
No heat absorbed by the calorimeter or container	Metal or plastic cup absorbs some heat
The reaction goes to completion instantly	Some reactions are slow or reversible
Temperature of the solution is uniform	Poor stirring can cause uneven heating

B. Common Sources of Error

Error Type	Explanation	Impact on ΔH
Heat lost to surroundings	Incomplete insulation, especially in neutralisation	ΔH too low (less exothermic)
Incomplete combustion (fuel)	Carbon may form instead of CO_2	ΔH too low
Evaporation of volatile substances	e.g. alcohols during weighing	Incorrect mass, inaccurate ΔH
Inaccurate temperature readings	Low-resolution thermometer	Under/overestimated ΔT
Heat absorbed by container	Not accounted for in $q = mc\Delta T$	q is underestimated

C. Improving Accuracy and Minimising Uncertainty

Method	Improvement
Use a lid on the cup	Minimises heat loss
Use a digital thermometer or datalogger	More precise temperature readings
Stir the solution constantly	Ensures uniform temperature
Use a bomb calorimeter (for combustion)	Minimises energy loss and ensures complete combustion
Use insulation (e.g., polystyrene cup in a beaker)	Reduces energy loss

D. Cooling Curve Corrections (for neutralisation or dissolution reactions)

Used when:

- Heat loss occurs before maximum temperature is reached.
- Need to estimate **maximum temperature rise** if no loss had occurred.

Method:

1. Record temperature **before, during, and after** the reaction.
2. Plot a **temperature vs time graph**.
3. Extrapolate **two lines**:
 - Pre-reaction temperature trend
 - Post-reaction cooling trend
4. The **intersection point** gives **maximum temperature rise** (ΔT used in $q = mc\Delta T$).

E. Commenting on Reliability vs Accuracy vs Precision

Term	Meaning	Example Comment
Accuracy	How close result is to true value	"Value is lower than data book value due to heat loss."
Precision	How close repeated results are to each other	"Results had a small range of values (± 0.1 °C), showing precision."
Reliability	Consistency of measurements across trials	"Repeat runs showed similar trends, increasing reliability."

F. Evaluating Experimental Enthalpy Values

If your calculated ΔH differs from data book:

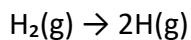
- Say **why**: "Heat loss to surroundings", "Incomplete combustion", "Evaporation", etc.
- Explain the **effect**: "This causes ΔH to be less exothermic than expected."
- Never just say: "Human error" — **always be specific**.

6.9 — Bond Enthalpy and Mean Bond Enthalpy

A. Definition of Bond Enthalpy

Bond Enthalpy (also called bond dissociation enthalpy)

The enthalpy change required to break **1 mole** of a **specific covalent bond** in a **gaseous molecule** under standard conditions.

Example:

$$\Delta H = +436 \text{ kJ mol}^{-1}$$

(This is the bond enthalpy of the H–H bond.)

- Always **endothermic** because energy is required to break bonds.

B. Definition of Mean Bond Enthalpy**Mean Bond Enthalpy**

The **average enthalpy change** required to break **1 mole** of a particular type of bond, averaged across **different compounds** containing that bond in the **gaseous state**.

Example:

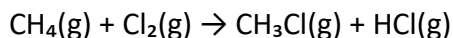
The **C–H** bond in CH_4 has a mean bond enthalpy of about $+412 \text{ kJ mol}^{-1}$, averaged over many organic molecules, not just methane.

- Still always **endothermic**
- Not exact for any single molecule, which introduces **limitations**

C. Calculating Enthalpy Change Using Bond Enthalpies**Hess-style method:**

$$\Delta H = \Sigma(\text{Bonds Broken}) - \Sigma(\text{Bonds Formed})$$

- Bonds broken = reactants (input energy, +)
- Bonds formed = products (release energy, –)

D. Example Question**Reaction:**

Given mean bond enthalpies (in kJ mol^{-1}):

- C–H = +412
- Cl–Cl = +243
- C–Cl = +338
- H–Cl = +431

Step 1: Bonds Broken (Reactants):

- $1 \times \text{C-H} = +412$
- $1 \times \text{Cl-Cl} = +243$
 $= +655 \text{ kJ mol}^{-1}$

Step 2: Bonds Formed (Products):

- $1 \times \text{C-Cl} = -338$
- $1 \times \text{H-Cl} = -431$
 $= -769 \text{ kJ mol}^{-1}$

$$\Delta H = 655 - 769 = -114 \text{ kJ mol}^{-1}$$

→ Exothermic reaction.

E. Limitations of Mean Bond Enthalpies

Limitation	Explanation
Mean values are averaged , not specific	Bond enthalpy values come from multiple molecules; not exact for any one molecule i.e. averaged across multiple molecules
All substances must be in the gaseous state	Bond enthalpy values do not include enthalpy changes of state
Assumes bonds are independent	Bond enthalpy may vary depending on molecular environment (e.g., C-H in CH_4 vs CH_3CH_3)
Not as accurate as $\Delta_f H^\circ$ / $\Delta_c H^\circ$ from data tables	For precise enthalpy values, use standard enthalpy data instead of bond enthalpies

F. When to Use Bond Enthalpy vs Hess Cycle

Use Bond Enthalpy When...	Use Hess Cycle When...
Only bond energies are provided	Standard enthalpy changes ($\Delta_f H^\circ$ or $\Delta_c H^\circ$) are provided
All substances involved are in the gaseous state	Includes liquids or solids (e.g., $\text{H}_2\text{O}(\text{l})$, $\text{CO}_2(\text{g})$)
Rough estimate is acceptable	Accurate thermodynamic value is required

6.10 — Calculating Mean Bond Enthalpies from Enthalpy Changes

A. What You Need to Do

If you're given:

- The **overall enthalpy change** (ΔH) of a reaction

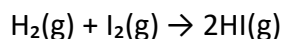
- The **bond enthalpies** of all other bonds
- You can calculate the **unknown bond enthalpy** using:

$$\Delta H = \Sigma(\text{Bonds Broken}) - \Sigma(\text{Bonds Formed})$$

$$\text{Bond Enthalpy (unknown)} = \Sigma(\text{Bonds Broken or Formed}) \pm \Delta H$$

B. Example Question

Reaction:



Given:

- $\Delta H = +53 \text{ kJ mol}^{-1}$
- Bond enthalpy H–H = $+436 \text{ kJ mol}^{-1}$
- Bond enthalpy I–I = $+151 \text{ kJ mol}^{-1}$
- Find mean bond enthalpy of H–I

Step 1: Bonds Broken (Reactants)

- $1 \times \text{H–H} = +436$
- $1 \times \text{I–I} = +151$
= **$+587 \text{ kJ mol}^{-1}$**

Step 2: Bonds Formed (Products)

Let H–I bond = x

$$2 \times \text{H–I} = 2x$$

Apply Hess formula:

$$\Delta H = \text{Bonds Broken} - \text{Bonds Formed}$$

$$+53 = 587 - 2x$$

$$2x = 587 - 53 = 534$$

$$x = 267 \text{ kJ mol}^{-1}$$

Answer: Mean bond enthalpy of H–I = **$+267 \text{ kJ mol}^{-1}$**

C. Notes on Method

- Works only if there is **one unknown**.
- Always express bond enthalpy as a **positive** value (energy required to break bonds).

- Final answer must include **units**: kJ mol^{-1} .

D. Exam Tips

Tip	Explanation
Check whether it's 1 mol or 2 mol of product	Adjust your calculation accordingly
Check if ΔH is given for 1 mol or multiple mol	e.g., $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$ means $4 \times \text{O-H}$ bonds
Ensure sign convention is correct	Breaking = +, Forming = -
Answer must be mean bond enthalpy not ΔH	Double-check the question wording

6.11 — Bond Enthalpies and Predicting Reactivity

A. Core Concept

The strength of bonds — indicated by their **bond enthalpies** — gives insight into:

- Which bonds will break first** in a reaction
- How easily or quickly** a reaction occurs
- Whether a reaction is likely to proceed at room temperature**

This is particularly useful when comparing **reactivity trends** in organic and inorganic chemistry.

B. Key Principle

Bonds with lower bond enthalpies break **more easily** and therefore react **faster** or at **lower temperatures**.

This is because:

- They require **less activation energy** to break.
- The **activation energy barrier** is smaller.

C. Example 1: Halogenoalkanes

C-I bond has lower bond enthalpy than **C-Br** or **C-Cl**

Bond enthalpies:

- $\text{C-Cl} \approx +338 \text{ kJ mol}^{-1}$
- $\text{C-Br} \approx +276 \text{ kJ mol}^{-1}$
- $\text{C-I} \approx +238 \text{ kJ mol}^{-1}$

Conclusion:

- C–I breaks most easily → **iodoalkanes react fastest** in nucleophilic substitution.
- This is confirmed by hydrolysis rate experiments.

D. Example 2: Diatomic Elements

Compare:

- H–H bond = +436 kJ mol⁻¹
- Cl–Cl bond = +243 kJ mol⁻¹
- Br–Br bond = +193 kJ mol⁻¹
- I–I bond = +151 kJ mol⁻¹

Conclusion:

- I₂ is most reactive in terms of bond-breaking (easiest to break).
- But overall reactivity of halogens **decreases down the group** due to other factors like electronegativity and atom size.

E. Application to Activation Energy (E_a)

Bond enthalpy can be used to estimate the **activation energy (E_a)** for a reaction:

- E_a ≈ Energy needed to **break initial bonds** (not total ΔH)
- Lower bond enthalpy → Lower E_a → **Faster reaction**

F. Exam Question Formats You Might See

1. **"Explain why compound A reacts faster than compound B in terms of bond enthalpies."**
→ Discuss bond enthalpy values and link to lower activation energy.
2. **"State which bond will break first and justify your answer."**
→ Identify the bond with the **lowest bond enthalpy**.
3. **"Predict which halogenoalkane hydrolyses fastest and explain."**
→ Relate to **C–X bond strength**, not the halogen's electronegativity.